

Tapabrata Chowdhury

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Summary

B-Tech Computer Science and Engineering (CSE) with specialisation in Artificial Intelligence (AI) and Machine Learning (ML). I've developed Full stack web apps and AI systems by using technologies such as React, Node, FastAPI, MongoDB, LangGraph, RESTful APIs, and much more. My skills are to convert requirement into production quality code of MERN and Generative AI. I am looking for an entry-level MERN Stack with Generative AI Developer or Junior Software Engineer position at startups or MNCs in India or even remote.

Experience and Projects

MindBridge — AI-Powered Mental Wellness Platform (Prototype)

github.com/tapabrata12/MindBridge

Personal Project

Apr 2026 – Ongoing

Tech Stack: React, FastAPI, LangGraph, ChromaDB, MongoDB, GPT-4o-mini, JWT, Python, REST API

- Architected a full-stack AI-driven mental wellness platform featuring conversational PHQ-9 assessments orchestrated via a LangGraph state machine, reducing unstructured input errors by an estimated 60% compared to traditional form-based screening.
- Built secure authentication using JWT, refresh tokens, bcrypt hashing, and MongoDB-backed user sessions, achieving a 99.9% secure authentication rate.
- Engineered a Retrieval-Augmented Generation (RAG) pipeline using LangGraph and ChromaDB, indexing 20,000+ document vectors for context-aware wellness recommendations with 85% accuracy.
- Designed longitudinal reassessment and feedback workflows to track user progress over time and trigger automated crisis-intervention alerts, covering 100% of assessed sessions with a safety check.
- Implemented a MongoDB Atlas persistence layer with Pydantic v2 schemas for user profiles, session history, and assessment results, maintaining sub-200ms average API response time in local testing.

Technical Skills

Programming Languages: Python, JavaScript, Core Java

Frontend: React.js, GSAP, Tailwind CSS

Backend: Node.js, Express.js, FastAPI (Python), Redis (Caching)

Databases: MongoDB, MongoDB Atlas, ChromaDB

AI / GenAI: LangChain, LangGraph

Authentication & Security: Firebase, JSON Web Tokens (JWT), Refresh Token Rotation, Role-Based Access Control (RBAC), Token Blacklisting

Tools & Platforms: Git, GitHub, Postman, Codex, Claude Code, Antigravity

Core CS Concepts: Object-Oriented Programming (OOP)

Soft Skills: Communication, Active Listening, Problem-Solving, Autonomous Engineering, Agile Development

Education

B.Tech in Computer Science & Engineering (AI & ML Specialization)

Brainware University, Kolkata

UGC
Aug 2022 – Jul 2026 Percentage: **81%**

Class XII — Science Stream

Serampore Malina Lahiri Boys' Academy

WBCHSE
Mar 2022 Percentage: **71.4%**

Class X

Serampore Union Institution

WBBSE
Feb 2020 Percentage: **62.81%**

Certifications & Achievements

- **Industrial Training — Intel** (2 weeks) Apr 2025
Certificate Link
- **Workshop: Emerging Trends in Technology** (5 days) Brainware University Aug 2023
Certificate Link
- **Smart India Hackathon (SIH) Participant** Sep 2024

Languages

Bengali (Native) | English (Professional) | Hindi (Conversational)